LOTUS INSTITUTE OF MANAGEMENT, BAREILLY
IMPORTANT QUESTIONS

Disclaimer: The Important Questions provided in this document are only for reference and practice purposes. Lotus College does not guarantee that these questions will appear in the University examination. Students are advised to prepare the complete University syllabus. The syllabus can be downloaded from University Website.

Course: B.C.A	**Subject Code:** BCA-201 N
Semester: 3rd	**Subject Name: Object oriented programming using java**

Section – A

In **Section A**, all questions carry **2 marks.** The students must answer each question in approximately **75 words.**

S.No.	Questions	Unit	Year
1	Define java.	1	-
2	Differentiate between JDK and JVM.	1	2025
3	What is a Java Applet ?	1	-
4	Explain the features of java and how it differs from C++.	1	2025
5	What is java Token?	1	2025
6	Define object -oriented programming.	2	2025
7	What is an operator?	2	-
8	what is exception handling.	2	2025
9	Define the Class.	2	-
10	What is the use of constructor in java.	2	2025
11	Explain the concept of method overloading ?	2	-
12	What is a static variable?	2	-
13	What is method nesting?	2	-
14	Why is main() declared static?	2	-
15	Define super keyword.	2	-
16	Write a short note on final methods.	3	2025
17	Explain abstract class.	3	-
18	Define an interface in Java.	3	2025
19	Define Array in java.	3	-
20	Write a short note on packages.	3	2025
21	Explain difference between == and equals() for Strings?	3	-
22	How is Vector different from an array?	3	-
23	What is a wrapper class?	3	2025

24	Define autoboxing.	3	-
25	Explain difference between primitive types and wrapper classes.	3	-
26	Differentiate between throw and throws keywords.	4	-
27	What is an error?	4	-
28	Define the keyword throw.	4	-
29	Which keywords are used for exception handling?	4	-
30	What is sleep() method?	4	-
31	Explain the concept of method overloading.	4	-
32	Name the types of errors in java.	4	-
33	What is the difference between start() and run()?	4	-
34	What is multithreading?	4	-
35	How can a thread be created in Java?	4	-
36	What is the Runnable interface?	5	-
37	What is yield() method?	5	-
38	What is the throws keyword?	5	-
39	What is thread synchronization?	5	2025
40	What is a synchronized block?	5	-
41	What is a stream?	5	2025
42	Explain the purpose of the file class in java	5	2025
43	What is the purpose of try-catch in file handling? suppositories.	5	-

Section – B

In **Section B**, all questions carry **10 marks.** The students must answer each question in approximately 200 **words.**

S.No.	Questions	Unit	Year
1	Why is java platform independent.	1	-
2	Discuss the inheritance in java with suitable examples.	1	2025
3	What is encapsulation?	1	-
4	Define String class and its methods.	2	-
5	What is encapsulation?	3	-
6	Explain overloading and overriding in java.	3	-
7	Explain thread class.	3	-
8	Write short note on wrapper class and package.	4	-
9	Describe various types of exceptions in java and explain how they handled using try catch and finally.	5	2025

10	Explain file handling and its methods in java.	5	-

Section – C

In **Section C**, all questions carry **7 marks.** The students must answer each question in approximately 800 **words.**

S.No.	Questions	Unit	Year
1	What is Java? Explain the important features of Java in detail.	1	-
2	Explain the concept of Object-Oriented Programming. Discuss encapsulation, inheritance, polymorphism and abstraction with examples.	1	2025
3	Differentiate between JVM, JRE and JDK. Explain their roles in Java programming.	1	-
4	Explain primitive and non-primitive data types in Java with examples.	1	-
5	Explain if, if-else, nested if, else-if ladder and switch statements with examples.	1	-
6	Explain the concepts of classes and objects in Java. How are objects created and accessed?	2	-
7	What is a constructor? Explain default and parameterized constructors with suitable examples.	2	-
8	What are static members? Explain static variables, static methods and static blocks with examples.	2	-
9	Explain different types of inheritance supported by Java with suitable examples	2	-
10	Differentiate between method overloading and method overriding	2	
11	What are abstract methods and abstract classes? Explain them with suitable examples.	2	-
12	What are interfaces in java? Explain inheritance. Write a program to demonstrate the implementation of interfaces.	3	2025
13	Differentiate between an interface and an abstract class.	3	-
14	Explain thread creation by extending the Thread class with a suitable program.	4	-
15	What is an array? Explain declaration, creation and initialization of arrays in Java.	4	-
16	Write a Java program to find the largest element in an array.	4	-
17	Explain the String class and its important methods such as length(), charAt(), substring() and equals().	4	-

18	What is a Vector? Explain its features and important methods with examples.	5	-
19	What are wrapper classes? Explain their need and importance with examples.	5	-
20	What in detail the concept of multithreading in java. Discuss thread creation, life cycle, and thread priority with suitable code examples.	5	2025
21	Explain the complete syntax of exception handling using try, catch and finally.	5	-
22	What is synchronization? Why is synchronization required in multithreaded programming?	5	-
23	What is synchronization? Why is synchronization required in multithreaded programming?	5	-
24	Write a Java program demonstrating multithreading and synchronization.	5	-
25	Describe the concept of Input/output streams in java. Explain in different classes used for byte and character stream handling with some examples.	6	2025
26	Explain how to create, delete and check the properties of a file using the File class.	6	-
27	Explain thread creation by extending the `Thread` class with a suitable program	5	-